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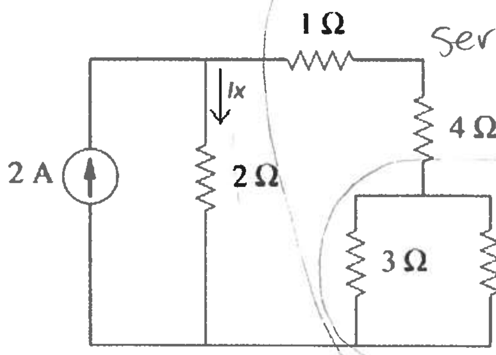
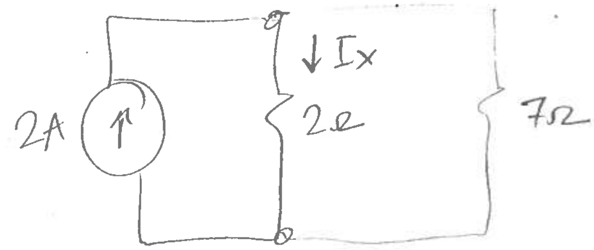
EE 2010 • Exam II • 15 September 2025 NAME SOLUTION

1) Use current division to determine I_x.



series; $\equiv 7\Omega$ +1

parallel; $\equiv 2\Omega$ +1



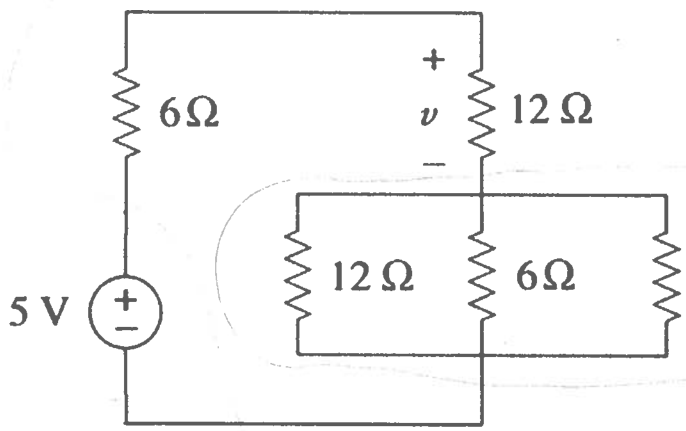
$$R_{eq.} = 2\Omega \parallel 7\Omega = \frac{14}{9}\Omega \quad +1$$

$$\therefore \; I_x = 2\left(\frac{14/9}{2}\right) \quad +1 \quad +1 \quad +1$$

$$I_x = 14/9 \text{ A} \quad +1$$

or 1.556 A

2) Use voltage division to find v.



$\equiv 3\Omega$ +1

$$R_{eq} = 6\Omega + 12\Omega + 3\Omega$$

$$= 21\Omega \quad +1$$

$$\therefore \; V = 5\left(\frac{12}{21}\right) = \frac{20}{7} \text{ V} \quad +1 \quad +1 \quad +1 \quad +1$$

or 2.85 V

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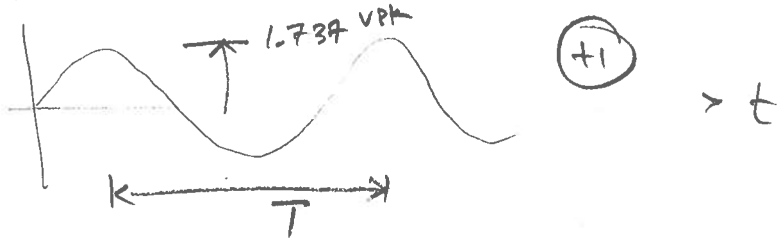
4) Sketch a 1.228 V_{RMS} sine wave with a frequency of 1 kHz, indicating the peak amplitude and period. Also sketch the current waveform that would result if this voltage was applied to a 600-Ω resistor. Determine the power in the resistor.

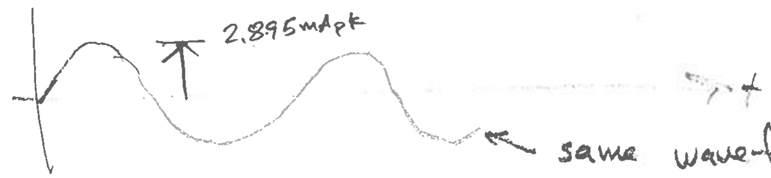
$V_{pk} = V_{RMS}\sqrt{2} = 1.228\sqrt{2} = 1.737\,V_{pk}$ (+1)

$T = \frac{1}{f} = \frac{1}{1k} = 1ms$ (+1)



(+1)

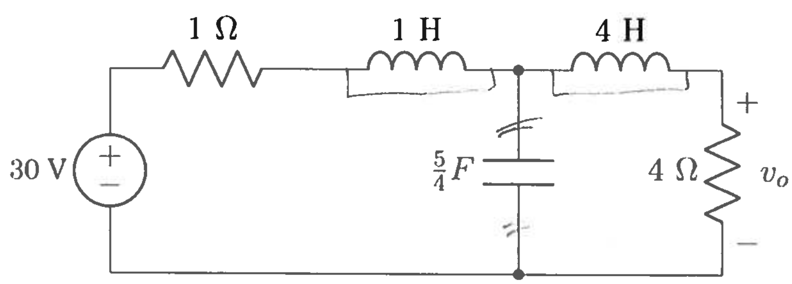
$I_{pk} = \frac{V_{pk}}{R} = \frac{1.737}{600} = 0.002895$ or $2.895\,mA_{pk}$ (+1)



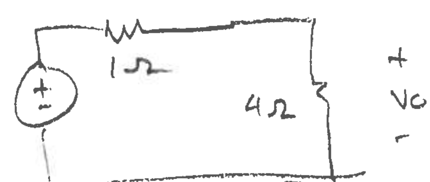
← same waveform (+1)

$P = \frac{V_{RMS}^2}{R} = \frac{1.228^2}{600} = 0.002513$ or $2.513\,mW$ (+1)

5) Determine v_o, assuming the current source is operating at DC. Hint: redraw the circuit.



at D.C. :



+2

Inductors → short
Capacitor → open

+2

$$V_o = 30\left(\frac{4}{1+4}\right)$$

$\therefore V_o = 24V$ (+1)